Status	Finished
Started	Thursday, 13 August 2026, 3:36 PM
Completed	Thursday, 13 August 2026, 3:44 PM
Duration	8 mins 22 secs
Marks	11.00/12.00
Grade	**9.17** out of 10.00 (**91.67**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which of the following is a **DISADVANTAGE** of Mesh topology?

a. Requires a central hub that becomes a single point of failure

b. A large amount of cabling and many I/O ports are required, making it expensive and hard to scale

c. Poor fault tolerance — one cable failure brings down the network

d. All devices must share the same frequency channel

e. Data collisions are very common due to shared access

The correct answer is: A large amount of cabling and many I/O ports are required, making it expensive and hard to scale

Question 2

Complete

Mark 1.00 out of 1.00

What is the CRITICAL disadvantage of Ring topology that can affect the entire network?

a. The central hub must be replaced every year

b. If any single node goes down, the entire network collapses

c. Only one device can ever be added to the network

d. Every device can see all data on the network, reducing security

e. Data can only travel at very low speeds

The correct answer is: If any single node goes down, the entire network collapses

Question 3

Complete

Mark 1.00 out of 1.00

In STAR topology, what is the PRIMARY role of the central hub or switch?

a. It assigns IP addresses to all connected devices

b. broadcasts data to all devices simultaneously

c. It encrypts all data before forwarding to protect privacy

d. It stores all data transmitted across the network

e. It acts as a connection junction, controlling and regulating all devices, and can also act as a repeater

The correct answer is: It acts as a connection junction, controlling and regulating all devices, and can also act as a repeater

Question 4

Complete

Mark 0.00 out of 1.00

A small startup needs a cost-effective network where a single device failure does NOT affect others, and devices can be easily added or removed. Which topology BEST fits?

a. Ring — most efficient data flow

b. Bus — cheapest cabling

c. Hybrid — combines all advantages

d. Mesh — best redundancy and security

e. Star — each device is independently connected; adding/removing is easy

The correct answer is: Star — each device is independently connected; adding/removing is easy

Question 5

Complete

Mark 1.00 out of 1.00

In BUS topology, all computers share a single cable. What happens if that central cable encounters a problem?

a. Only the devices nearest to the break are affected

b. Devices switch to wireless communication automatically

c. The entire network collapses — all communication stops

d. Data is automatically rerouted through backup cables

e. The hub automatically disconnects the broken segment

The correct answer is: The entire network collapses — all communication stops

Question 6

Complete

Mark 1.00 out of 1.00

What is the CORRECT distinction between 'data' and 'signal' in network communication?

- a. Data is what we understand (information representation); signal is what we can measure (energy movement in a medium)
- b. Signal is stored in databases; data travels over cables
- c. Signal is the high-level application content; data is the low-level electrical form
- d. Data and signal are exactly the same thing — both refer to bits
- e. Data is continuous; signal is always discrete

The correct answer is: Data is what we understand (information representation); signal is what we can measure (energy movement in a medium)

Question 7

Complete

Mark 1.00 out of 1.00

What is the primary responsibility of the physical layer?

- a. Carry bits across a physical link using suitable signals
- b. Forward packets across several interconnected network systems
- c. Deliver segments between application processes on different hosts
- d. Manage dialogs and synchronization between application sessions
- e. Translate application data into standardized presentation formats

The correct answer is: Carry bits across a physical link using suitable signals

Question 8

Complete

Mark 1.00 out of 1.00

The Physical Layer carries data as signals over transmission media. Which set of signal types does it use?

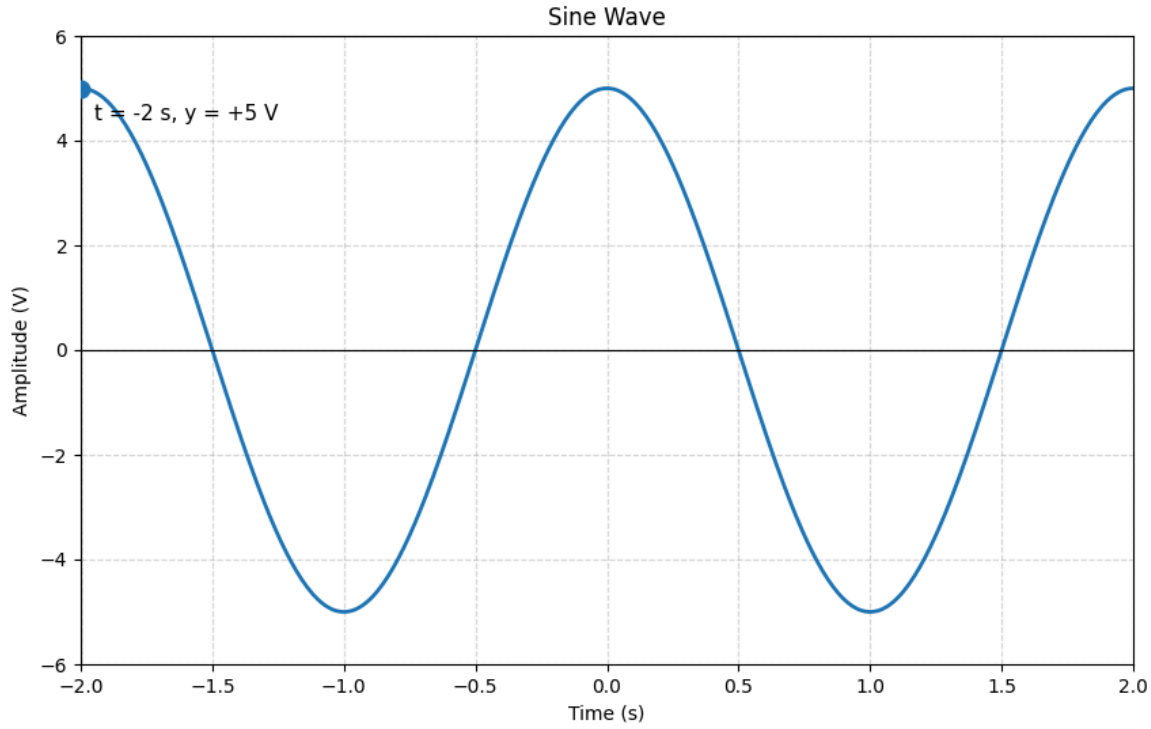
- a. IP packets and MAC frames
- b. Electrical signals, light signals (fiber optics), and radio waves (wireless)
- c. ASCII characters and binary code
- d. HTML pages and JSON data
- e. Segments and datagrams

The correct answer is: Electrical signals, light signals (fiber optics), and radio waves (wireless)

Question 9

Complete

Mark 3.00 out of 3.00



Determine the amplitude (A), frequency (f), and phase (ϕ) in degrees (0-360) of the sine wave above.

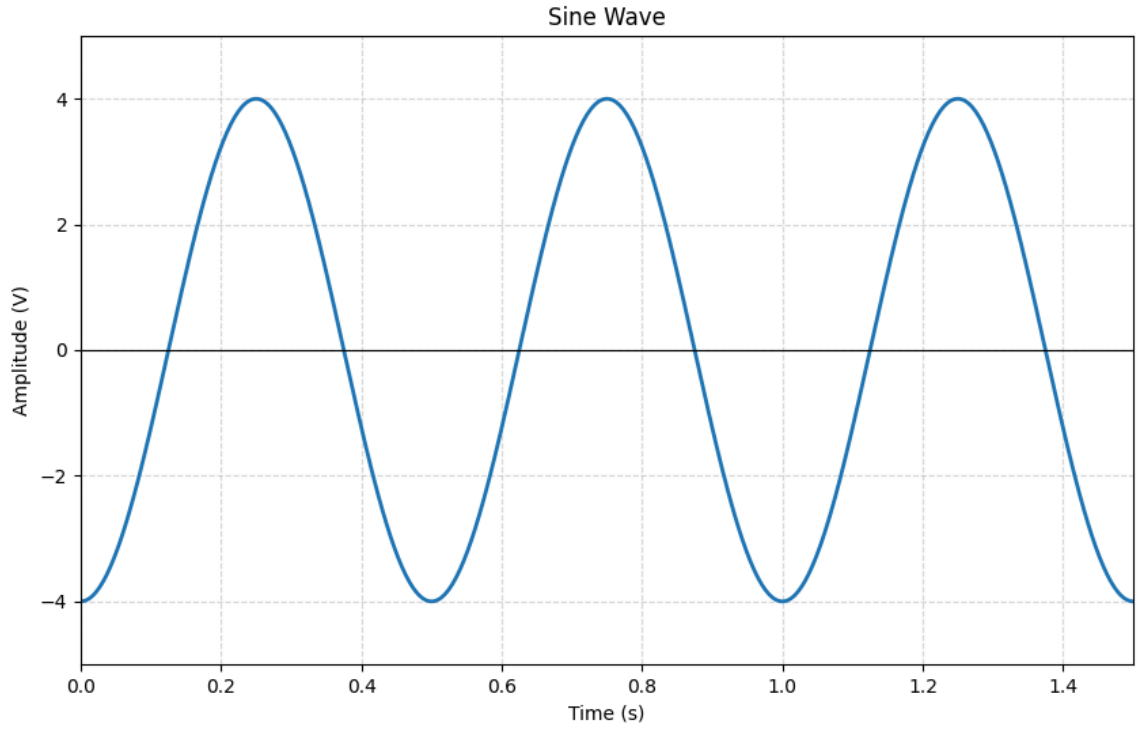
Answer: A = 5 V, f = 0.5 Hz, ϕ = 90 degrees.

T = 2 s --> f = 1/2 = 0.5 Hz

Question 10

Complete

Mark 1.00 out of 1.00



Which equation represents the signal above?

a. $y(t) = 4\sin\left(4\pi t - \frac{\pi}{2}\right)$

b. $y(t) = 4\sin\left(2\pi t + \frac{\pi}{2}\right)$

c. $y(t) = 4\sin\left(4\pi t + \frac{\pi}{2}\right)$

d. $y(t) = 4\sin\left(2\pi t - \frac{\pi}{2}\right)$

e. $y(t) = 4\sin\left(6\pi t - \frac{\pi}{2}\right)$

The signal completes 3 cycles in 1.5 s, so f = 3/1.5 = 2 Hz

The correct answer is:

$$y(t) = 4\sin\left(4\pi t - \frac{\pi}{2}\right)$$